

Functions & Graphs

ANSWER KEY



Evaluating functions

- 1 If $f(x) = 3x^2 - 2x + 5$, find $f(-2)$.
- $$f(-2) = 3(4) - 2(-2) + 5 = 12 + 4 + 5 = 21.$$
- 2 If $g(x) = \frac{2x+1}{x-3}$, find $g(5)$.
- $$g(5) = \frac{2(5)+1}{5-3} = \frac{11}{2}.$$

Domain and range

- 3 Find the domain of $f(x) = \frac{4}{x+7}$.
- The denominator cannot equal zero: $x + 7 \neq 0$, so $x \neq -7$. Domain: all reals except $x = -7$.
- 4 Find the domain of $g(x) = \sqrt{2x-8}$.
- The radicand must be non-negative: $2x - 8 \geq 0$, so $x \geq 4$. Domain: $x \geq 4$.

Composite functions

- 5 If $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $f(g(2))$.
- First, $g(2) = 4 - 3 = 1$. Then $f(1) = 2(1) + 1 = 3$.
- 6 If $f(x) = x + 4$ and $g(x) = 3x$, find $(f \circ g)(x)$ and $(g \circ f)(x)$, and explain why they differ.
- $(f \circ g)(x) = f(3x) = 3x + 4$. $(g \circ f)(x) = g(x + 4) = 3(x + 4) = 3x + 12$. They differ because function composition is generally not commutative — the order in which functions are applied changes the result.

Graphing linear and quadratic functions

- 7 Find the x - and y -intercepts of $y = 2x - 6$.
- y -intercept (set $x = 0$): $y = -6$, so $(0, -6)$. x -intercept (set $y = 0$): $0 = 2x - 6$, so $x = 3$, giving $(3, 0)$.
- 8 Find the vertex of $y = x^2 - 8x + 15$.
- The vertex's x -coordinate is $-\frac{b}{2a} = -\frac{-8}{2} = 4$. $y = 16 - 32 + 15 = -1$. Vertex: $(4, -1)$.

Transformations of graphs

- 9 The graph of $y = x^2$ is shifted 4 units left and 3 units down. Write the equation of the resulting graph.

Shifting left by 4 replaces x with $x + 4$; shifting down 3 subtracts 3: $y = (x + 4)^2 - 3$.

- 10 Describe how the graph of $y = -3(x - 2)^2 + 1$ differs from the graph of $y = x^2$.

The graph is reflected over the x -axis and vertically stretched by a factor of 3 (from the -3), shifted 2 units right (from $x - 2$), and shifted 1 unit up (from $+1$).

Piecewise and step functions

- 11 A function is defined as $f(x) = 2x$ for $x \leq 2$ and $f(x) = x + 5$ for $x > 2$. Find $f(0)$ and $f(4)$.

Since $0 \leq 2$, use the first piece: $f(0) = 2(0) = 0$. Since $4 > 2$, use the second piece: $f(4) = 4 + 5 = 9$.

Interpreting graphs in context

- 12 A graph shows a car's speed over time: the speed increases from 0 to 40 mph over the first 10 seconds, stays constant at 40 mph for the next 20 seconds, then decreases to 0 over the final 10 seconds. Describe what is happening to the car during each phase.

During the first phase (0-10 seconds), the car is accelerating from a stop up to 40 mph. During the second phase (10-30 seconds), the car travels at a constant speed of 40 mph (cruising). During the third phase (30-40 seconds), the car is decelerating, coming to a stop.

Finding the equation of a function from a graph description

- 13 A parabola has x -intercepts at $x = -1$ and $x = 5$, and passes through $(2, -9)$. Find its equation in factored form.

Factored form: $y = a(x + 1)(x - 5)$. Substitute $(2, -9)$: $-9 = a(3)(-3) = -9a$, so $a = 1$. Equation: $y = (x + 1)(x - 5)$.

Multi-part function analysis

- 14 This question has two parts. A function is given by $f(x) = x^2 - 4x - 5$. (a) Find the zeros of the function by factoring. (b) Find the vertex, and state whether it is a minimum or maximum.

(a) Factor: $x^2 - 4x - 5 = (x - 5)(x + 1) = 0$, giving zeros at $x = 5$ and $x = -1$. (b) Vertex x -coordinate: $-\frac{-4}{2} = 2$. $f(2) = 4 - 8 - 5 = -9$. Vertex: $(2, -9)$, which is a minimum since the parabola opens upward (positive leading coefficient).

Inverse functions

- 15 Find the inverse of $f(x) = 4x + 8$.

Let $y = 4x + 8$. Swap x and y : $x = 4y + 8$. Solve for y : $x - 8 = 4y$, so $y = \frac{x - 8}{4}$. Thus $f^{-1}(x) = \frac{x - 8}{4}$.

- 16 If $f(3) = 10$, find $f^{-1}(10)$, using the definition of an inverse function.

By definition, $f^{-1}(10) = 3$, since inverse functions reverse input-output pairs.

Rational and radical functions

- 17 Find the vertical asymptote of $f(x) = \frac{3}{x - 5}$.

The vertical asymptote occurs where the denominator equals zero: $x - 5 = 0$, so $x = 5$.

- 18 Find $g(9)$ for $g(x) = \sqrt{x} + 4$.

$$g(9) = \sqrt{9} + 4 = 3 + 4 = 7.$$

Exponential functions

- 19 If $f(x) = 2^x$, find $f(5)$ and $f(-2)$.

$$f(5) = 2^5 = 32. \quad f(-2) = 2^{-2} = \frac{1}{4}.$$

- 20 A function is given by $g(x) = 100(0.5)^x$. Find $g(3)$ and interpret its meaning if this models a decaying quantity over x time periods.

$g(3) = 100(0.5)^3 = 100(0.125) = 12.5$. This means after 3 time periods, the original quantity of 100 has decayed to 12.5 — consistent with the quantity halving each period.

Multi-part analysis of a real-world function model

- 21 This question has two parts. A ball is thrown, and its height is modeled by $h(t) = -16t^2 + 48t + 4$, where t is time in seconds. (a) Find the height of the ball at $t = 0$ and interpret this value. (b) Find the time at which the ball reaches its maximum height, and find that maximum height.

(a) $h(0) = 4$, meaning the ball starts at a height of 4 feet (perhaps thrown from a platform or by someone's hand). (b) Maximum occurs at $t = -\frac{48}{2(-16)} = 1.5$ seconds.
 $h(1.5) = -16(2.25) + 48(1.5) + 4 = -36 + 72 + 4 = 40$ feet.

Multiple choice — easy

22 If $f(x) = -2x^2 + 7x + 5$, find $f(-3)$.

a. -34 ✓

b. -36

c. -32

d. -44

$$f(-3) = -2(-3)^2 + 7(-3) + 5 = -34.$$

23 If $f(x) = 4x^2 - 9x - 6$, find $f(1)$.

a. -9

b. -11 ✓

c. -13

d. 1

$$f(1) = 4(1)^2 - 9(1) - 6 = -11.$$

24 If $f(x) = -2x^2 + 4x + 9$, find $f(4)$.

a. -5

b. -9

c. -25

d. -7 ✓

$$f(4) = -2(4)^2 + 4(4) + 9 = -7.$$

25 If $f(x) = -3x^2 + 8x + 4$, find $f(6)$.

a. -54

b. -56 ✓

c. -58

d. -64

$$f(6) = -3(6)^2 + 8(6) + 4 = -56.$$

26 If $f(x) = 4x^2 + 9x - 4$, find $f(6)$.

a. 202

b. 194 ✓

c. 196

d. 192

$$f(6) = 4(6)^2 + 9(6) - 4 = 194.$$

27 If $f(x) = 2x^2 - 2x + 2$, find $f(-1)$.

a. 2

b. 6 ✓

c. 4

d. 8

$$f(-1) = 2(-1)^2 - 2(-1) + 2 = 6.$$

28 If $f(x) = -3x^2 - 7x - 6$, find $f(1)$.

a. -4

b. -18

c. -14

d. -16 ✓

$$f(1) = -3(1)^2 - 7(1) - 6 = -16.$$

29 If $f(x) = 3x^2 + 3x - 1$, find $f(-1)$.

a. -1 ✓

b. 1

c. 4

d. -3

$$f(-1) = 3(-1)^2 + 3(-1) - 1 = -1.$$

30 If $f(x) = 3x^2 + 5$, find $f(2)$.

a. 15

b. 7

c. 19

d. 17 ✓

$$f(2) = 3(2)^2 + 0(2) + 5 = 17.$$

31 If $f(x) = -x^2 - 9x + 4$, find $f(1)$.

a. -4

b. -14

c. -6 ✓

d. -8

$$f(1) = -(1)^2 - 9(1) + 4 = -6.$$

32 If $f(x) = x^2 - 6x + 3$, find $f(2)$.

a. -5 ✓

b. -3

c. -11

d. -7

$$f(2) = (2)^2 - 6(2) + 3 = -5.$$

33 If $f(x) = -x^2 - 8x + 6$, find $f(-2)$.

a. 20

b. 18 ✓

c. 6

d. 16

$$f(-2) = -(-2)^2 - 8(-2) + 6 = 18.$$

34 If $f(x) = -x^2 + x + 8$, find $f(-4)$.

a. -12 ✓

b. -28

c. -10

d. -14

$$f(-4) = -(-4)^2 + 1(-4) + 8 = -12.$$

35 If $f(x) = 3x^2 - 9$, find $f(-3)$.

a. 36

b. 18 ✓

c. 20

d. 16

$$f(-3) = 3(-3)^2 + 0(-3) - 9 = 18.$$

36 If $f(x) = x^2 - 5x - 6$, find $f(-4)$.

- a. 42
- b. 32
- c. 30 ✓
- d. 28

$$f(-4) = (-4)^2 - 5(-4) - 6 = 30.$$

Multiple choice — medium

37 Find the vertex of $y = -3x^2 + 42x - 134$.

- a. (7, 13) ✓
- b. (7, -13)
- c. (-7, 13)
- d. (8, 13)

The vertex's x-coordinate is $-\frac{b}{2a} = -\frac{42}{-6} = 7$. $y = -3(7)^2 + 42(7) - 134 = 13$. Vertex: (7, 13).

38 Find the vertex of $y = 3x^2 + 36x + 107$.

- a. (-5, -1)
- b. (6, -1)
- c. (-6, 1)
- d. (-6, -1) ✓

The vertex's x-coordinate is $-\frac{b}{2a} = -\frac{36}{6} = -6$. $y = 3(-6)^2 + 36(-6) + 107 = -1$. Vertex: (-6, -1).

39 Find the vertex of $y = 3x^2 - 42x + 141$.

- a. (7, 6)
- b. (8, -6)
- c. (7, -6) ✓
- d. (-7, -6)

The vertex's x-coordinate is $-\frac{b}{2a} = -\frac{(-42)}{6} = 7$. $y = 3(7)^2 - 42(7) + 141 = -6$. Vertex: (7, -6).

40 Find the vertex of $y = -2x^2 - 4x + 4$.

- a. (-1, -6)
- b. (0, 6)
- c. (-1, 6) ✓
- d. (1, 6)

The vertex's x-coordinate is $-\frac{b}{2a} = -\frac{(-4)}{-4} = -1$. $y = -2(-1)^2 - 4(-1) + 4 = 6$. Vertex: (-1, 6).

41 Find the vertex of $y = -3x^2 + 12x - 20$.

- a. (3, -8)
- b. (2, -8) ✓
- c. (2, 8)
- d. (-2, -8)

The vertex's x-coordinate is $-\frac{b}{2a} = -\frac{12}{-6} = 2$. $y = -3(2)^2 + 12(2) - 20 = -8$. Vertex: (2, -8).

42 Find the vertex of $y = x^2 - 14x + 46$.

- a. (7, 3)
- b. (8, -3)
- c. (-7, -3)
- d. (7, -3) ✓

The vertex's x-coordinate is $-\frac{b}{2a} = -\frac{(-14)}{2} = 7$. $y = (7)^2 - 14(7) + 46 = -3$. Vertex: (7, -3).

43 Find the vertex of $y = -x^2 + 16x - 60$.

a. (9, 4)

b. (8, 4) ✓

c. (-8, 4)

d. (8, -4)

The vertex's x -coordinate is $-\frac{b}{2a} = -\frac{16}{-2} = 8$. $y = -(8)^2 + 16(8) - 60 = 4$. Vertex: (8, 4).

44 Find the vertex of $y = -x^2 - 2x - 13$.

a. (0, -12)

b. (-1, 12)

c. (-1, -12) ✓

d. (1, -12)

The vertex's x -coordinate is $-\frac{b}{2a} = -\frac{(-2)}{-2} = -1$. $y = -(-1)^2 - 2(-1) - 13 = -12$. Vertex: (-1, -12).

45 Find the vertex of $y = 3x^2 + 48x + 195$.

a. (8, 3)

b. (-8, -3)

c. (-8, 3) ✓

d. (-7, 3)

The vertex's x -coordinate is $-\frac{b}{2a} = -\frac{48}{6} = -8$. $y = 3(-8)^2 + 48(-8) + 195 = 3$. Vertex: (-8, 3).

46 Find the vertex of $y = -3x^2 - 42x - 161$.

a. $(-7, -14)$ ✓

b. $(-7, 14)$

c. $(7, -14)$

d. $(-6, -14)$

The vertex's x-coordinate is $-\frac{b}{2a} = -\frac{(-42)}{-6} = -7$. $y = -3(-7)^2 - 42(-7) - 161 = -14$. Vertex: $(-7, -14)$.

47 Find the vertex of $y = -3x^2 - 6x + 9$.

a. $(-1, -12)$

b. $(-1, 12)$ ✓

c. $(1, 12)$

d. $(0, 12)$

The vertex's x-coordinate is $-\frac{b}{2a} = -\frac{(-6)}{-6} = -1$. $y = -3(-1)^2 - 6(-1) + 9 = 12$. Vertex: $(-1, 12)$.

48 Find the vertex of $y = -x^2 - 8x - 3$.

a. $(4, 13)$

b. $(-3, 13)$

c. $(-4, -13)$

d. $(-4, 13)$ ✓

The vertex's x-coordinate is $-\frac{b}{2a} = -\frac{(-8)}{-2} = -4$. $y = -(-4)^2 - 8(-4) - 3 = 13$. Vertex: $(-4, 13)$.

49 Find the vertex of $y = -3x^2 + 42x - 136$.

a. $(7, -11)$

b. $(7, 11)$ ✓

c. $(-7, 11)$

d. $(8, 11)$

The vertex's x -coordinate is $-\frac{b}{2a} = -\frac{42}{-6} = 7$. $y = -3(7)^2 + 42(7) - 136 = 11$. Vertex: $(7, 11)$.

50 Find the vertex of $y = 2x^2 + 16x + 45$.

a. $(-4, -13)$

b. $(-4, 13)$ ✓

c. $(-3, 13)$

d. $(4, 13)$

The vertex's x -coordinate is $-\frac{b}{2a} = -\frac{16}{4} = -4$. $y = 2(-4)^2 + 16(-4) + 45 = 13$. Vertex: $(-4, 13)$.

51 Find the vertex of $y = -x^2 - 2x - 1$.

a. $(1, 0)$

b. $(-3, 1)$

c. $(0, 0)$

d. $(-1, 0)$ ✓

The vertex's x -coordinate is $-\frac{b}{2a} = -\frac{(-2)}{-2} = -1$. $y = -(-1)^2 - 2(-1) - 1 = 0$. Vertex: $(-1, 0)$.

Multiple choice — hard

52 If $f(x) = -x + 4$ and $g(x) = 3x + 9$, find $f(g(0))$.

- a. 9
- b. -7
- c. -5 ✓
- d. -3

First, $g(0) = 3(0) + 9 = 9$. Then $f(9) = -(9) + 4 = -5$.

53 If $f(x) = 4x + 5$ and $g(x) = 2x + 6$, find $f(g(4))$.

- a. 63
- b. 61 ✓
- c. 59
- d. 14

First, $g(4) = 2(4) + 6 = 14$. Then $f(14) = 4(14) + 5 = 61$.

54 If $f(x) = -x - 8$ and $g(x) = -3x + 1$, find $f(g(2))$.

- a. -3 ✓
- b. -7
- c. -5
- d. -1

First, $g(2) = -3(2) + 1 = -5$. Then $f(-5) = -(-5) - 8 = -3$.

55 If $f(x) = 2x - 4$ and $g(x) = -3x - 5$, find $f(g(0))$.

- a. -14 ✓
- b. -5
- c. -16
- d. -12

First, $g(0) = -3(0) - 5 = -5$. Then $f(-5) = 2(-5) - 4 = -14$.

56 If $f(x) = -4x + 8$ and $g(x) = -x + 1$, find $f(g(-1))$.

a. -2

b. -3

c. 0 ✓

d. 2

First, $g(-1) = -(-1) + 1 = 2$. Then $f(2) = -4(2) + 8 = 0$.

57 Find the inverse of $f(x) = 3x + 9$.

a. $\frac{x+9}{3}$

b. $\frac{x-7}{3}$

c. $\frac{x-9}{3}$ ✓

d. $\frac{x-11}{3}$

Let $y = 3x + 9$. Swap x and y : $x = 3y + 9$. Solving for y gives $f^{-1}(x) = \frac{x-9}{3}$.

58 Find the inverse of $f(x) = -5x + 11$.

a. $\frac{x-9}{-5}$

b. $\frac{x-11}{-5}$ ✓

c. $\frac{x+11}{-5}$

d. $\frac{x-13}{-5}$

Let $y = -5x + 11$. Swap x and y : $x = -5y + 11$. Solving for y gives $f^{-1}(x) = \frac{x-11}{-5}$.

59 Find the inverse of $f(x) = 4x + 11$.

- a. $\frac{x-9}{4}$
- b. $\frac{x+11}{4}$
- c. $\frac{x-13}{4}$
- d. $\frac{x-11}{4}$ ✓

Let $y = 4x + 11$. Swap x and y : $x = 4y + 11$. Solving for y gives $f^{-1}(x) = \frac{x-11}{4}$.

60 Find the inverse of $f(x) = 2x - 1$.

- a. $\frac{x+1}{2}$ ✓
- b. $\frac{x-1}{2}$
- c. $\frac{x+3}{2}$
- d. $\frac{x+1}{3}$

Let $y = 2x - 1$. Swap x and y : $x = 2y - 1$. Solving for y gives $f^{-1}(x) = \frac{x+1}{2}$.

61 Find the inverse of $f(x) = 5x - 3$.

- a. $\frac{x+5}{5}$
- b. $\frac{x-3}{5}$
- c. $\frac{x+1}{5}$
- d. $\frac{x+3}{5}$ ✓

Let $y = 5x - 3$. Swap x and y : $x = 5y - 3$. Solving for y gives $f^{-1}(x) = \frac{x+3}{5}$.

62 If $f(x) = 5 \cdot 3^x$, find $f(2)$.

- a. 30
- b. 48
- c. 45 ✓
- d. 42

$$f(2) = 5 \cdot 3^2 = 5(9) = 45.$$

63 If $f(x) = 2 \cdot 3^x$, find $f(3)$.

a. 18

b. 54 ✓

c. 57

d. 51

$$f(3) = 2 \cdot 3^3 = 2(27) = 54.$$

64 If $f(x) = 2 \cdot 4^x$, find $f(4)$.

a. 516

b. 508

c. 32

d. 512 ✓

$$f(4) = 2 \cdot 4^4 = 2(256) = 512.$$

65 If $f(x) = 4 \cdot 4^x$, find $f(-2)$.

a. $\frac{1}{8}$

b. 1

c. -32

d. $\frac{1}{4}$ ✓

$$f(-2) = 4 \cdot 4^{-2} = 4 \cdot \frac{1}{4^2} = 4/16 = \frac{1}{4}.$$

66 If $f(x) = 4 \cdot 4^x$, find $f(3)$.

a. 252

b. 48

c. 256 ✓

d. 260

$$f(3) = 4 \cdot 4^3 = 4(64) = 256.$$